So far in this section, we have been asking what is the least order of a complete graph G such that every 2-colouring of its edges yields a monochromatic copy of some given graph H. Rather than keeping G complete and focusing on its order, let us now consider its structure too, i.e., minimize G with respect to the subgraph relation. Given a graph H, let us call a graph G *Ramsey-minimal* for H if G is minimal with the property that every 2-colouring of its edges yields a monochromatic copy of H.

Ramsey-minimal

What do such Ramsey-minimal graphs look like? Are they unique? The following result, which we include for its pretty proof, answers the second question for some H:

Proposition 9.2.3. *If T is a tree but not a star, then infinitely many graphs are Ramsey-minimal for T.*

(1.5.3)
(5.2.3)
(5.2.5)
(=11.2.2)

Proof. Let $|T| =: r$. We show that for every $n \in \mathbb{N}$ there is a graph of order at least n that is Ramsey-minimal for T.

By Theorem 5.2.5, there exists a graph G with chromatic number $\chi(G) > r^2$ and girth $g(G) > n$. If we colour the edges of G red and green, then the red and the green subgraph cannot both have an r-(vertex-)colouring in the sense of Chapter 5: otherwise we could colour the vertices of G with the pairs of colours from those colourings and obtain a contradiction to $\chi(G) > r^2$. So let $G' \subseteq G$ be monochromatic with $\chi(G') > r$. By Lemma 5.2.3, G' has a subgraph of minimum degree at least r, which contains a copy of T by Corollary 1.5.3.

Let $G^* \subseteq G$ be Ramsey-minimal for T. Clearly, G^* is not a forest: the edges of any forest can be 2-coloured (partitioned) so that no monochromatic subforest contains a path of length 3, let alone a copy of T. (Here we use that T is not a star, and hence contains a P^3.) So G^* contains a cycle, which has length $g(G) > n$ since $G^* \subseteq G$. In particular, $|G^*| > n$ as desired. □

9.3 Induced Ramsey theorems

Ramsey's theorem can be rephrased as follows. For every graph $H = K^r$ there *exists* a graph G such that every 2-colouring of the edges of G yields a monochromatic $H \subseteq G$. This is witnessed by any large enough complete graph as G. Let us now change the problem slightly and ask for a graph G in which every 2-edge-colouring yields a monochromatic *induced* $H \subseteq G$, where H is now an arbitrary given graph.

This slight modification changes the character of the problem. What is needed now is no longer a simple proof that G is 'big enough' (as for Theorem 9.1.1), but a careful construction: the construction of a graph that, however we bipartition its edge set, contains an induced copy of H

with all edges in one partition class. We shall call such a graph a *Ramsey graph* for H. *Ramsey graph*

The fact that every graph H has such a Ramsey graph G was proved around 1973, independently by Deuber, by Erdős, Hajnal & Pósa, and by Rödl. It is one of the fundamental results of graph Ramsey theory.

Theorem 9.3.1. *Every graph has a Ramsey graph. In other words, for every graph H there exists a graph G which, for every partition*[4] *$\{E_1, E_2\}$ of $E(G)$, contains H as an induced subgraph with $E(H) \subseteq E_1$ or $E(H) \subseteq E_2$.*

We shall prove Theorem 9.3.1 for bipartite graphs first, and use this in our proof of the general theorem. For the remainder of this section, let us view bipartite graphs P as triples (V_1, V_2, E), where V_1 and V_2 are the two vertex classes and $E \subseteq V_1 \times V_2$ is the set of edges. The reason for this more explicit notation is that we want embeddings between bipartite graphs to respect their bipartitions: given another bipartite graph $P' = (V_1', V_2', E')$, an injective map $\varphi\colon V_1 \cup V_2 \to V_1' \cup V_2'$ will be called an *embedding* of P in P' if $\varphi(V_i) \subseteq V_i'$ for $i = 1, 2$ and $\varphi(v_1)\varphi(v_2)$ is an edge of P' if and only if $v_1 v_2$ is an edge of P. Note that such embeddings are 'induced'. Instead of $\varphi\colon V_1 \cup V_2 \to V_1' \cup V_2'$ we may simply write $\varphi\colon P \to P'$. *bipartite* *embedding* $P \to P'$

Recall that $[X]^k$ denotes the set of k-subsets of a set X. $[X]^k$

Lemma 9.3.2. *Every bipartite graph can be embedded in a bipartite graph of the form $(X, [X]^k, E)$ with $E = \{\, xY \mid x \in Y \,\}$.* E

Proof. Let P be any bipartite graph, with vertex classes $\{a_1, \dots, a_n\}$ and $\{b_1, \dots, b_m\}$, say. Let X be a set with $2n + m$ elements, say

$$X = \{x_1, \dots, x_n,\ y_1, \dots, y_n,\ z_1, \dots, z_m\}\,;$$

we shall define an embedding $\varphi\colon P \to (X, [X]^{n+1}, E)$.

Let us start by setting $\varphi(a_i) := x_i$ for all $i = 1, \dots, n$. Which $(n+1)$-sets $Y \subseteq X$ are suitable candidates for the choice of $\varphi(b_i)$ for a given vertex b_i? Clearly those adjacent exactly to the images of the neighbours of b_i, i.e. those satisfying

$$Y \cap \{x_1, \dots, x_n\} = \varphi(N_P(b_i))\,. \tag{1}$$

Since $d(b_i) \leqslant n$, the requirement of (1) leaves at least one of the $n+1$ elements of Y unspecified. In addition to $\varphi(N_P(b_i))$, we may therefore include in each $Y = \varphi(b_i)$ the vertex z_i as an 'index'; this ensures that $\varphi(b_i) \neq \varphi(b_j)$ for $i \neq j$, even when b_i and b_j have the same neighbours in P. To specify the sets $Y = \varphi(b_i)$ completely, we finally fill them up with 'dummy' elements y_j until $|Y| = n + 1$. □

[4] The partition classes here are allowed to be empty.

Next, we prove that every bipartite graph has a Ramsey graph – even a bipartite one:

Lemma 9.3.3. *For every bipartite graph P there exists a bipartite graph P' such that for every 2-colouring of the edges of P' there is an embedding $\varphi\colon P \to P'$ for which all the edges of $\varphi(P)$ have the same colour.*

(9.1.3) P, X, k, E P', X', k' *Proof.* We may assume by Lemma 9.3.2 that P has the form $(X, [X]^k, E)$ with $E = \{\, xY \mid x \in Y \,\}$. We show the assertion for the graph $P' := (X', [X']^{k'}, E')$, where $k' := 2k-1$, X' is any set of cardinality

$$|X'| = R\left(k',\, 2\tbinom{k'}{k},\, k\,|X| + k - 1\right),$$

(this is the Ramsey number defined after Theorem 9.1.3), and

E'
$$E' := \{\, x'Y' \mid x' \in Y' \,\}\,.$$

α, β Let us then colour the edges of P' with two colours α and β. Of the $|Y'| = 2k-1$ edges incident with a vertex $Y' \in [X']^{k'}$, at least k must have the same colour. For each Y' we may therefore choose a fixed k-set Z' $Z' \subseteq Y'$ such that all the edges $x'Y'$ with $x' \in Z'$ have the same colour; *associated* we shall call this colour *associated* with Y'.

The sets Z' can lie within their supersets Y' in $\binom{k'}{k}$ ways, as follows. Let X' be linearly ordered. Then for every $Y' \in [X']^{k'}$ there is a unique $\sigma_{Y'}$ order-preserving bijection $\sigma_{Y'}\colon Y' \to \{1, \dots, k'\}$, which maps Z' to one of $\binom{k'}{k}$ possible images.

We now colour $[X']^{k'}$ with the $2\binom{k'}{k}$ elements of the set

$$[\{1, \dots, k'\}]^k \times \{\alpha, \beta\}$$

as colours, giving each $Y' \in [X']^{k'}$ as its colour the pair $(\sigma_{Y'}(Z'), \gamma)$, where γ is the colour α or β associated with Y'. Since $|X'|$ was chosen as the Ramsey number with parameters k', $2\binom{k'}{k}$ and $k\,|X| + k - 1$, we W know that X' has a monochromatic subset W of cardinality $k\,|X| + k - 1$. All Z' with $Y' \subseteq W$ thus lie within their Y' in the same way, i.e. there exists an $S \in [\{1, \dots, k'\}]^k$ such that $\sigma_{Y'}(Z') = S$ for all $Y' \in [W]^{k'}$, and α all $Y' \in [W]^{k'}$ are associated with the same colour, say with α.

$\varphi|_X$ x_i, w_i, n We now construct the desired embedding φ of P in P'. We first define φ on $X =: \{x_1, \dots, x_n\}$, choosing images $\varphi(x_i) =: w_i \in W$ so that $w_i < w_j$ in our ordering of X' whenever $i < j$. Moreover, we choose the w_i so that exactly $k-1$ elements of W are smaller than w_1, exactly $k-1$ lie between w_i and w_{i+1} for $i = 1, \dots, n-1$, and exactly $k-1$ are bigger than w_n. Since $|W| = kn + k - 1$, this can indeed be done (Fig. 9.3.1).

$\varphi|_{[X]^k}$ We now define φ on $[X]^k$. Given $Y \in [X]^k$, we wish to choose $\varphi(Y) =: Y' \in [X']^{k'}$ so that the neighbours of Y' among the vertices

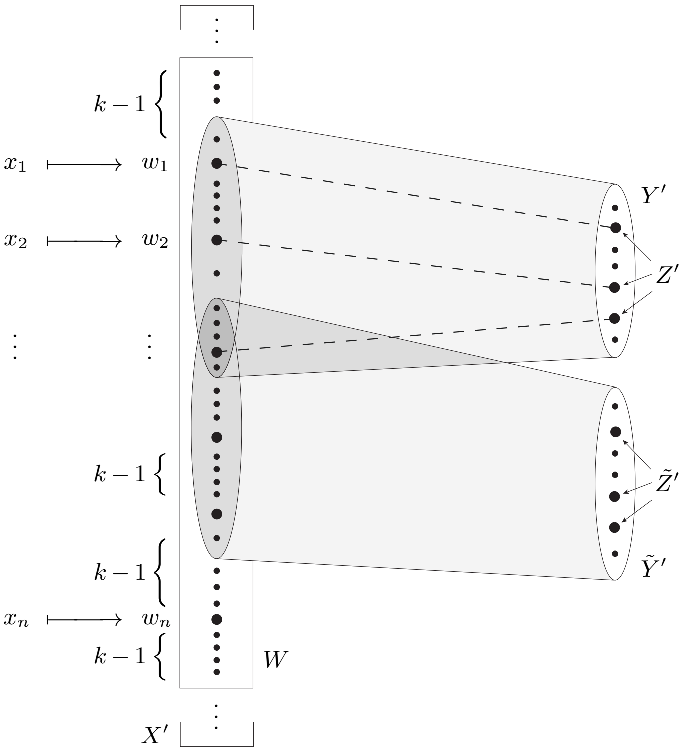

Fig. 9.3.1. The graph of Lemma 9.3.3

in $\varphi(X)$ are precisely the images of the neighbours of Y in P, i.e. the k vertices $\varphi(x)$ with $x \in Y$, and so that all these edges at Y' are coloured α. To find such a set Y', we first fix its subset Z' as $\{\, \varphi(x) \mid x \in Y \,\}$ (these are k vertices of type w_i) and then extend Z' by $k' - k$ further vertices $u \in W \smallsetminus \varphi(X)$ to a set $Y' \in [W]^{k'}$, in such a way that Z' lies correctly within Y', i.e. so that $\sigma_{Y'}(Z') = S$. This can be done, because $k - 1 = k' - k$ other vertices of W lie between any two w_i. Then

$$Y' \cap \varphi(X) = Z' = \{\, \varphi(x) \mid x \in Y \,\},$$

so Y' has the correct neighbours in $\varphi(X)$, and all the edges between Y' and these neighbours are coloured α (because those neighbours lie in Z' and Y' is associated with α). Finally, φ is injective on $[X]^k$: the images Y' of different vertices Y are distinct, because their intersections with $\varphi(X)$ differ. Hence, our map φ is indeed an embedding of P in P'. □

Proof of Theorem 9.3.1. Let H be given as in the theorem, and let $n := R(r)$ be the Ramsey number of $r := |H|$. Then, for every 2-colouring r, n

K of its edges, the graph $K = K^n$ contains a monochromatic copy of H – although not necessarily induced.

We start by constructing a graph G^0, as follows. Imagine the vertices of K to be arranged in a column, and replace every vertex by a row of $\binom{n}{r}$ vertices. Then each of the $\binom{n}{r}$ columns arising can be associated with one of the $\binom{n}{r}$ ways of embedding $V(H)$ in $V(K)$; let us furnish this column with the edges of such a copy of H. The graph G^0 thus arising consists of $\binom{n}{r}$ disjoint copies of H and $(n-r)\binom{n}{r}$ isolated vertices (Fig. 9.3.2).

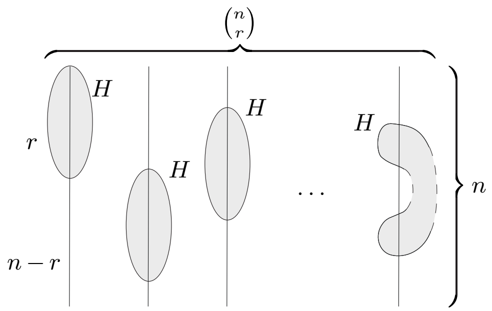

Fig. 9.3.2. The graph G^0

In order to define G^0 formally, we assume that $V(K) = \{1, \dots, n\}$ and choose copies $H_1, \dots, H_{\binom{n}{r}}$ of H in K with pairwise distinct vertex sets. (Thus, on each r-set in $V(K)$ we have one fixed copy H_j of H.) We then define

$$V(G^0) := \left\{ (i,j) \mid i = 1, \dots, n;\ j = 1, \dots, \binom{n}{r} \right\}$$

G^0 and

$$E(G^0) := \bigcup_{j=1}^{\binom{n}{r}} \left\{ (i,j)(i',j) \mid ii' \in E(H_j) \right\}.$$

The idea of the proof now is as follows. Our aim is to reduce the general case of the theorem to the bipartite case dealt with in Lemma 9.3.3. Applying the lemma iteratively to all the pairs of rows of G^0, we construct a very large graph G such that for every edge colouring of G there is an induced copy of G^0 in G that is monochromatic on all the bipartite subgraphs induced by its pairs of rows, i.e. in which edges between the same two rows always have the same colour. The projection of this $G^0 \subseteq G$ to $\{1, \dots, n\}$ (by contracting its rows) then defines an edge colouring of K. (If the contraction does not yield all the edges of K, colour the missing edges arbitrarily.) By the choice of $|K|$, some $K^r \subseteq K$ will be monochromatic. The H_j inside this K^r then occurs with the same colouring in the jth column of our G^0, where it is an induced subgraph of G^0, and hence of G.

Formally, we shall define a sequence $G^0, \ldots, G^m$ of n-partite graphs G^k, with n-partition $\{V_1^k, \ldots, V_n^k\}$ say, and then let $G := G^m$. The graph G^0 has been defined above; let $V_1^0, \ldots, V_n^0$ be its rows:

$$V_i^0 := \{ (i,j) \mid j = 1, \ldots, \tbinom{n}{r} \}.$$ V_i^0

Now let $e_1, \ldots, e_m$ be an enumeration of the edges of K. For $k = 0, \ldots, m-1$, construct G^{k+1} from G^k as follows. If $e_{k+1} = i_1 i_2$, say, let $P = (V_{i_1}^k, V_{i_2}^k, E)$ be the bipartite subgraph of G^k induced by its i_1th and i_2th row. By Lemma 9.3.3, P has a (minimal) bipartite Ramsey graph $P' = (W_1, W_2, E')$. We wish to define $G^{k+1} \supseteq P'$ in such a way that every (monochromatic) embedding $P \to P'$ can be extended to an embedding $G^k \to G^{k+1}$ respecting their n-partitions. Let $\{\varphi_1, \ldots, \varphi_q\}$ be the set of all embeddings of P in P', and let e_k, m i_1, i_2 P P' W_1, W_2 φ_p, q

$$V(G^{k+1}) := V_1^{k+1} \cup \ldots \cup V_n^{k+1},$$

where

$$V_i^{k+1} := \begin{cases} W_1 & \text{for } i = i_1 \\ W_2 & \text{for } i = i_2 \\ \bigcup_{p=1}^{q} (V_i^k \times \{p\}) & \text{for } i \notin \{i_1, i_2\}. \end{cases}$$

(Thus for $i \neq i_1, i_2$, we take as V_i^{k+1} just q disjoint copies of V_i^k.) We now define the edge set of G^{k+1} so that the obvious extensions of φ_p to all of $V(G^k)$ become embeddings of G^k in G^{k+1}: for $p = 1, \ldots, q$, let $\psi_p \colon V(G^k) \to V(G^{k+1})$ be defined by

$$\psi_p(v) := \begin{cases} \varphi_p(v) & \text{for } v \in P \\ (v, p) & \text{for } v \notin P \end{cases}$$

and let

$$E(G^{k+1}) := \bigcup_{p=1}^{q} \{ \psi_p(v)\psi_p(v') \mid vv' \in E(G^k) \}.$$

Now for every 2-colouring of its edges, G^{k+1} contains an induced copy $\psi_p(G^k)$ of G^k whose edges in P, i.e. those between its i_1th and i_2th row, have the same colour: just choose p so that $\varphi_p(P)$ is the monochromatic induced copy of P in P' that exists by Lemma 9.3.3.

We claim that $G := G^m$ satisfies the assertion of the theorem. So let a 2-colouring of the edges of G be given. By the construction of G^m from G^{m-1}, we can find in G^m an induced copy of G^{m-1} such that for $e_m = ii'$ all edges between the ith and the i'th row have the same colour. In the same way, we find inside this copy of G^{m-1} an induced copy of G^{m-2} whose edges between the ith and the i'th row have the same colour also for $ii' = e_{m-1}$. Continuing in this way, we finally arrive at an induced copy of G^0 in G such that, for each pair (i, i'), all the edges between V_i^0 and $V_{i'}^0$ have the same colour. As shown earlier, this G^0 contains a monochromatic induced copy H_j of H. □

The Ramsey graph G for H built in the proof of Theorem 9.3.1 is sparse in the sense that it contains no complete graphs bigger than it has to in order to contain H: G and H have the same clique number $\omega(G) = \omega(H)$ (Exercise 23). This property, which comes for free with this proof of the theorem, is non-trivial even if we do not ask for induced embeddings: the existence of a graph G that contains a monochromatic copy of H, induced or not, on any 2-colouring of its edges is neither obvious nor does it follow easily from Ramsey's original theorem.

An even stronger sparsity requirement on G is to ask that it should contain no shorter cycles than H does, that $g(G) = g(H)$. As a small indication of how ambitious this requirement on G is, notice that the statement of $\chi(G) > r$ can be rephrased as a Ramsey-like property of G: for any *vertex*-colouring with at most r colours, G contains a 'monochromatic' copy of $H = K^2$, an edge joining two like-coloured vertices. But as we noted in Chapter 5.2, graphs of simultaneously large chromatic number and girth are difficult to construct.

However, Ramsey graphs of large girth do exist:

Theorem 9.3.4. (Reiher & Rödl 2023)
Every graph H has a Ramsey graph G that satisfies $g(G) = g(H)$ unless H is a forest, in which case $g(G)$ can be made arbitrarily large.

If H is a tree but not a star, the aim of $g(G) = g(H)$ $(= \infty)$ cannot be realised: if $g(G) = \infty$ then G is a forest, so its edges can be 2-coloured alternately so that every path of length 3 receives two colours.

9.4 Ramsey properties and connectivity

According to Ramsey's theorem, every large enough graph G has a very dense or a very sparse induced subgraph of given order, a K^r or $\overline{K^r}$. If we assume that G is connected, we can say a little more:

Proposition 9.4.1. *For every integer $r \geqslant 1$ there is an $n \in \mathbb{N}$ such that every connected graph of order at least n contains K^r, $K_{1,r}$ or P^r as an induced subgraph.*

(1.3.3) *Proof.* Let $d+1$ be the Ramsey number of r, let $n \geqslant \frac{d}{d-2}(d-1)^r$, and let G be a graph of order at least n. If G has a vertex v of degree at least